5. **What is the bond order of CO group?**

(a) 1 (b) 2.5

(c) 3.5 (d) 3

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

In Molecular orbital theory the bond order formula can be defined as half of the difference between the number of electrons in bonding orbitals and anti-bonding orbitals.

Bond order = 1/2 [N_b-N_a]

Where,

N_b = number of bonding electrons

N_a = number of anti-bonding electrons

CO (Carbon monoxide) molecule has 10 valance electrons, four from carbon atom ($2s^2 2p^2$) and six from oxygen atom ($2s^2 2p^4$). According to molecular orbital diagram, molecular orbital configuration is given as

$\delta\, 2s^2\, \delta\, {}^*2s^2\, \delta\, 2pz^2\, \pi\, 2px^2\, \pi\, 2py^2$

here, N_b = 8; N_a = 2

Thus, Bond order of CO = 1/2[8 – 2] = 3

Note : Here the calculations are done ignoring the 1s orbitals because there will be 2 electrons in the bonding and 2 in the anti-bonding.

B. Hydrogen and its Compounds

1. **An element X has four electrons in its outermost orbit. What will be the formula of its compounds with Hydrogen?**

(a) X_4H (b) X_4H_4

(c) XH_3 (d) XH_4

43rd B.P.S.C. (Pre) 1999

Ans. (d)

The compounds of element X which has four electrons in its outermost orbit will be XH_4.

2. **Burning of hydrogen produces –**

(a) Oxygen (b) Ash

(c) Soil (d) Water

47th B.P.S.C. (Pre) 2005

Ans. (d)

Hydrogen gas is highly flammable and burns in air at a very wide range of concentrations between 4% to 75% by volume. Hydrogen gas cannot burn in absence of air. But by burning with oxygen it produces water.

3. **The chemical formula for heavy water is :**

(a) H_2O (b) N_2O

(c) D_2O (d) CuO

(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (c)

Heavy water (D_2O) or deuterium oxide is used as a moderator as well as a coolant in nuclear reactors because it slows down neutrons effectively and also has a low probability of absorption of neutrons. Deuterium is an isotope of hydrogen which comprises both a neutron and a proton. Deuterium or heavy hydrogen reacts with oxygen to form deuterium oxide (D_2O), also known as heavy water. Normal water (H_2O) also used as a moderator as well as a coolant in nuclear reactors.

4. **Heavy water is that water –**

(a) The temperature of which is kept constant at 4°C

(b) In which insoluble salts of Calcium and Potassium are present

(c) In which isotopes takes place of Hydrogen

(d) In which isotopes takes place of Oxygen

41st B.P.S.C. (Pre) 1996

Ans. (c)

See the explanation of the above question.

5. **Permanent hardness of water is due to –**

(a) Chlorides and sulphates of Calcium and Magnesium

(b) Calcium bicarbonate sulphates

(c) Magnesium bicarbonate

(d) Chlorides of Silver and Potassium

40th B.P.S.C. (Pre) 1995

Ans. (a)

Hard water is described as 'hard' due to the presence of highly dissolved minerals specifically sulphates and chlorides of calcium and magnesium. Hard water is salty and therefore not used for drinking. It is very difficult to wash clothes with hard water as it requires more soap and leaves a messy scum that cannot be washed out easily. Hard water blocks the Xylem tissues of the plants and thus not suitable for irrigation.

6. **The pH value of water is**

(a) 4 (b) 7

(c) 12 (d) 18

(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

65th B.P.S.C. (Pre) 2019

Ans. (b)

The pH value of pure water is 7.